SWIFTLOG LOGISTICS LIMITED

Data Analytics & Power BI Technical Project Documentation

AnalystLab Africa - Week 5 | SwiftLog Logistics Supply Chain Analytics & Business Intelligence

Purpose

This document consolidates the technical evidence for data import and quality checks, Power Query transformations, data modeling, DAX/KPI development, dashboard features, business analysis, limitations, and strategic recommendations. It complements the 3-5 page Executive BI Report and the executive presentation.

Document item	Reference
Project	SwiftLog Logistics Analytics & AI Transformation - Data Analytics Track
Reporting period	1 January 2026 - 8 September 2026
Primary tool	Microsoft Power BI Desktop
Primary model file	SwiftLog_Analytics(2).pbix
Assignment basis	AnalystLab Africa Experience Lab - Week 5 Official Weekly Assignment, Data Analytics Track, Parts 1-7

1. Data Import, Understanding & Preparation

All 11 operational CSV datasets were imported and profiled before modeling. Checks covered schema, data types, missing values, exact duplicates, key uniqueness, and foreign-key coverage. The source data dictionary was also reviewed as a reference file.

Dataset	Rows	Columns	Missing cells	Exact duplicates
Customers	1000	3	0	0
Orders	5000	4	0	0
Order Items	15000	4	0	0
Products	250	3	0	0
Suppliers	150	2	0	0
Warehouses	15	2	0	0
Inventory	4000	3	0	3
Shipments	5000	4	0	0
Delivery Performance	5000	3	0	0
Employees	200	2	0	0
Returns	500	3	0	0

Data-quality result

No missing values were found in the 11 operational CSV datasets. The only exact duplicates identified were three duplicate Inventory records; these were removed, reducing Inventory from 4,000 to 3,997 rows.

Duplicate Inventory combination	Action
W02 - P023 - Stock 207	One duplicate removed
W05 - P010 - Stock 404	One duplicate removed
W15 - P021 - Stock 218	One duplicate removed

Key / relationship check	Result
Orders.CustomerID -> Customers.CustomerID	100% matched
Order_Items.OrderID -> Orders.OrderID	100% matched
Order_Items.ProductID -> Products.ProductID	100% matched
Inventory.ProductID -> Products.ProductID	100% matched
Inventory.WarehouseID -> Warehouses.WarehouseID	100% matched
Shipments.OrderID -> Orders.OrderID	100% matched
Shipments.WarehouseID -> Warehouses.WarehouseID	100% matched
Returns.OrderID -> Orders.OrderID	100% matched
Employees.WarehouseID -> Warehouses.WarehouseID	100% matched

Key / relationship check	Result
Delivery_Performance.ShipmentID -> Shipments.ShipmentID	100% matched

Order dates span 1 January 2026 to 8 September 2026. September is therefore a partial month and is treated carefully in trend interpretation.

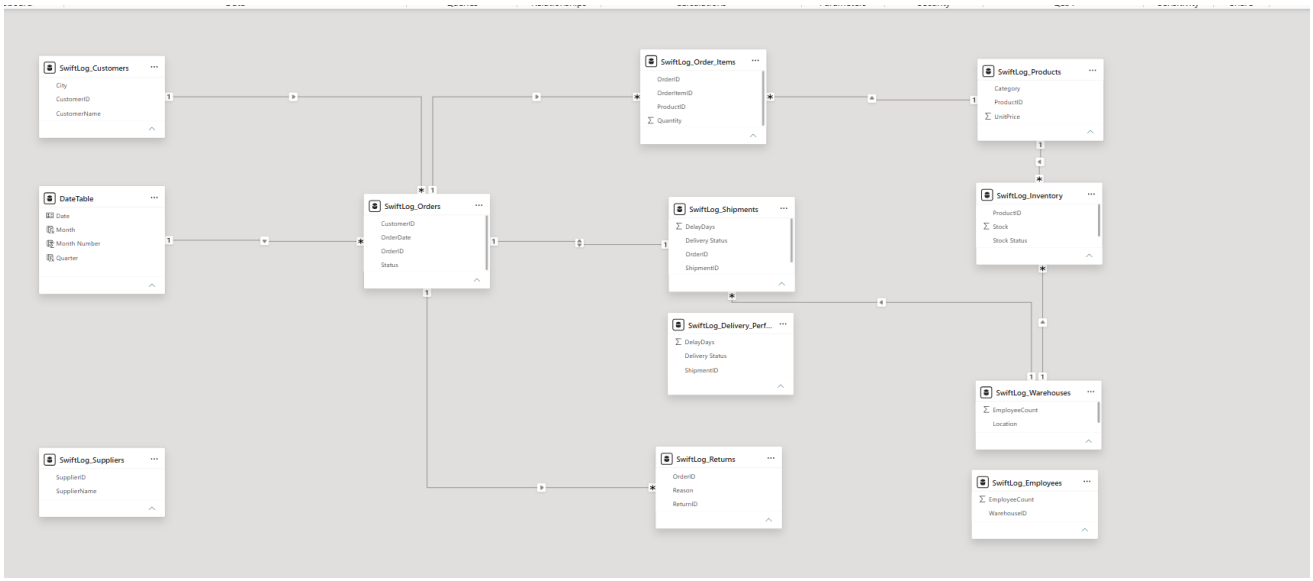


Figure 1. Power BI model view showing the main relationship structure.

1.1 Model design and relationships

Relationship	Cardinality / role
Customers[CustomerID] -> Orders[CustomerID]	1:* customer-to-order
DateTable[Date] -> Orders[OrderDate]	1:* date filtering
Orders[OrderID] -> Order_Items[OrderID]	1:* order-to-line items
Products[ProductID] -> Order_Items[ProductID]	1:* product sales
Products[ProductID] -> Inventory[ProductID]	1:* product inventory
Warehouses[WarehouseID] -> Inventory[WarehouseID]	1:* warehouse inventory
Warehouses[WarehouseID] -> Shipments[WarehouseID]	1:* warehouse shipment activity
Orders[OrderID] -> Shipments[OrderID]	1:1 in the source data
Orders[OrderID] -> Returns[OrderID]	1:* return records

Model limitation - Suppliers

The Suppliers table contains SupplierID and SupplierName, but no SupplierID is present in Products, Shipments, Inventory, Orders, or another operational table. Supplier performance therefore cannot be validly linked or scored from the supplied data. The table remains disconnected rather than forcing an artificial relationship.

2. Data Cleaning & Transformation (Power Query)

Cleaning was applied consistently across tables, with transformations chosen only where they improved data quality or the analytical model. The main steps are documented below, as required by the project guide.

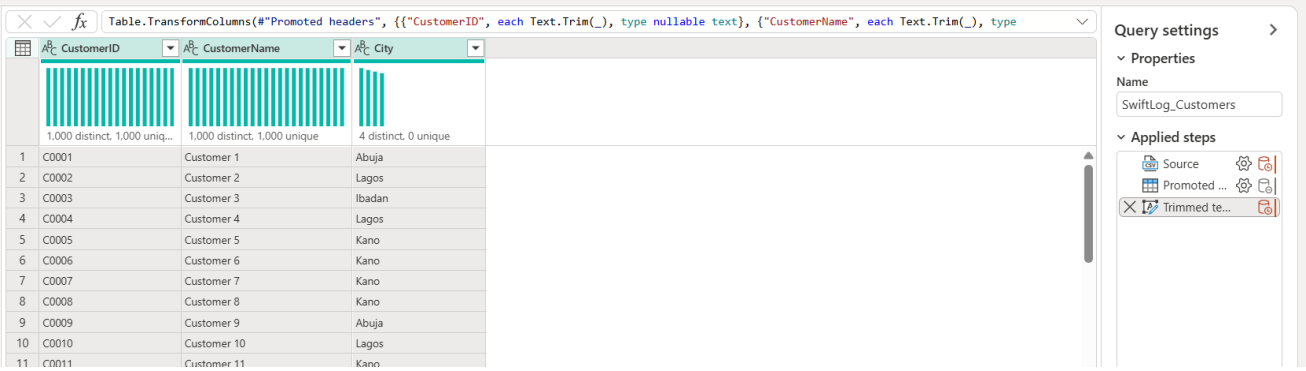


Figure 2. Power Query example: text standardization/trim step with Applied Steps visible.

Transformation	Tables / scope	Purpose
Promote first row as headers	Imported CSV tables	Ensure correct field names
Assign data types	All tables	IDs as Text; dates as Date; Quantity/Stock/DelayDays as Whole Number; UnitPrice as Decimal
Trim/Clean text	Text fields	Remove leading/trailing or non-printing text inconsistencies
Duplicate check and removal	Inventory	Removed 3 exact duplicate records after verification
Date table	Model-level DAX table	Provide controlled Month/Quarter sorting and time filtering
Month sort	DateTable[Month] by DateTable[Month Number]	Ensure chronological month order
Stock Status	Inventory	Business field for low-stock monitoring
Employee aggregation	Employees -> Warehouses	Grouped EmployeeID count by WarehouseID and merged as EmployeeCount
Delivery status integration	Delivery_Performance -> Shipments	Merged Status by ShipmentID and renamed to Delivery Status
Support-query approach	Employees and Delivery Performance	Source tables support transformations; integrated fields are used in core reporting tables

2.1 Dedicated Date table

A dedicated DateTable was created instead of relying only on Power BI automatic date hierarchies. Because the data covers a single year, the report uses Date, Month, Month Number, and Quarter as the primary reporting fields.

```
DateTable =
ADDCOLUMNS(
    CALENDAR(MIN(SwiftLog_Orders[OrderDate]), MAX(SwiftLog_Orders[OrderDate])),
    "Month", FORMAT([Date], "MMMM"),
    "Month Number", MONTH([Date]),
    "Quarter", "Q" & FORMAT([Date], "Q")
)
```

The Date column was marked as the table date field, and Month was sorted by Month Number.

2.2 Additional calculated/business fields

Field	Location	Definition / use
Stock Status	Inventory	Conditional business label used to identify low-stock inventory in dashboard visuals.
EmployeeCount	Warehouses	Count of employees grouped by WarehouseID and merged into the Warehouses table.
Delivery Status	Shipments	Status merged from Delivery Performance by ShipmentID (On Time / Delayed).
Month / Month Number / Quarter	DateTable	Time-intelligence and chronological sorting fields.

3. KPI Development & DAX Measures

The PBIX contains more than the minimum 10 custom measures required by the guide. The table below documents the measure intent and business value. DAX shown is the implemented/used analytical logic where it was verified from the model design and dashboard behavior.

Measure	Logic	Business value
Total Revenue	Delivered-order revenue	Core realized-sales KPI; excludes pending/cancelled orders.
Total Orders	Distinct order count	Tracks business activity and demand volume.
Total Customers	Distinct customer count	Shows customer base size.
Total Products	Distinct product count	Shows product portfolio size.
Total Shipments	Distinct shipment count	Tracks logistics workload.
Average Order Value	Revenue per qualifying order	Supports customer/commercial value analysis.
Average Delivery Delay	Average DelayDays for delivered orders	Quantifies delivery lateness severity.
On-Time Delivery Rate	Share of delivered orders with Delivery Status = On Time	Primary delivery reliability KPI.
Delayed Shipments	Count of delayed shipment-status records	Highlights operational exception volume.
Inventory Stock Level	Sum of Stock	Monitors total inventory held.
Units Ordered	Sum of Order_Items[Quantity]	Measures product demand volume.

Measure	Logic	Business value
Inventory Turnover	Units Ordered / Inventory Stock Level (proxy)	Relative demand-to-stock indicator; not formal accounting turnover.
Low Stock Products	Distinct low-stock products	Prioritizes replenishment.
Return Rate	Returns relative to orders	Monitors service/quality issues.
Total Warehouses	Distinct warehouse count	Network footprint KPI.
Selected Product	SELECTEDVALUE(ProductID)	Supports drill-through/tooltip context.
Dynamic Executive Trends Title	Selected month-aware title text	Improves contextual storytelling.

3.1 Representative DAX

```
Total Revenue =
CALCULATE(
    SUMX(
        SwiftLog_Order_Items,
        SwiftLog_Order_Items[Quantity] * RELATED(SwiftLog_Products[UnitPrice])
    ),
    SwiftLog_Orders[Status] = "Delivered"
)

Total Orders = DISTINCTCOUNT(SwiftLog_Orders[OrderID])
Total Customers = DISTINCTCOUNT(SwiftLog_Customers[CustomerID])
Total Products = DISTINCTCOUNT(SwiftLog_Products[ProductID])
Total Shipments = DISTINCTCOUNT(SwiftLog_Shipments[ShipmentID])
Inventory Stock Level = SUM(SwiftLog_Inventory[Stock])
Units Ordered = SUM(SwiftLog_Order_Items[Quantity])

Average Delivery Delay =
CALCULATE(
    AVERAGE(SwiftLog_Shipments[DelayDays]),
    SwiftLog_Orders[Status] = "Delivered"
)

On-Time Delivery Rate =
DIVIDE(
    CALCULATE(
        [Total Shipments],
        SwiftLog_Shipments[Delivery Status] = "On Time",
        SwiftLog_Orders[Status] = "Delivered"
    ),
    CALCULATE([Total Shipments], SwiftLog_Orders[Status] = "Delivered"),
    0
)

Inventory Turnover = DIVIDE([Units Ordered], [Inventory Stock Level], 0)

Selected Product = SELECTEDVALUE(SwiftLog_Products[ProductID], "Select a Product")

Dynamic Executive Trends Title =
VAR SelectedMonth = SELECTEDVALUE(DateTable[Month], "All Months")
RETURN "Executive Trends - " & SelectedMonth
```

Important metric note

Formal Inventory Turnover normally requires cost of goods sold and average inventory. Those fields are not supplied, so the report uses a clearly documented demand-to-stock proxy. Likewise, Average Delivery Time cannot be calculated because shipment/delivery timestamps are not available; Average Delivery Delay is used instead.

4. Dashboard Development & Advanced Features

The final PBIX contains eight report pages: six main analytical pages plus a hidden Product Details drill-through page and a hidden Product Tooltip page. The report definition also confirms a Reset Filters bookmark, synchronized Month slicers, a dynamic title measure, page navigation, mobile layouts, and enhanced tooltips.

Page	Primary purpose
Executive Dashboard	Executive KPI cards and global filters
Executive Trends	Revenue, orders, delivery and return trends
Customer Analytics	Customer distribution, growth, order frequency and top customers
Inventory & Warehouse Analytics	Inventory, low stock, turnover proxy and warehouse comparison
Shipment & Delivery Analytics	Shipment trend, delay, status and regional performance
Product & Supplier Analytics	Product revenue, category, best-selling products, return reasons and supplier limitation
Product Details (hidden)	ProductID drill-through with revenue, units, inventory and monthly trend
Product Tooltip (hidden)	Hover tooltip showing selected product, revenue, units ordered and stock

Advanced feature	Implementation evidence
Drill-through	Product Details page bound to Products[ProductID] and hidden from normal navigation
Bookmarks	Reset Filters bookmark
Dynamic tooltip	Hidden Product Tooltip page (320 x 240) used by product visuals
Page navigation	Page Navigator visuals across report pages
Synchronized filters	Month slicer sync group used across selected pages
Cross-filtering	Interactive visuals use default report cross-filter/drill behavior
Dynamic title	Dynamic Executive Trends Title measure responds to Month selection
Conditional formatting	Used where appropriate in analytical tables/visuals
Advanced DAX	Business measures listed in Section 3
Mobile layout	Mobile definitions are present for main report visuals
Theme/branding	Consistent blue Power BI theme, page titles and card styling
What-If parameter	Not forced; guide states “where appropriate” and no defensible scenario was required for the descriptive model

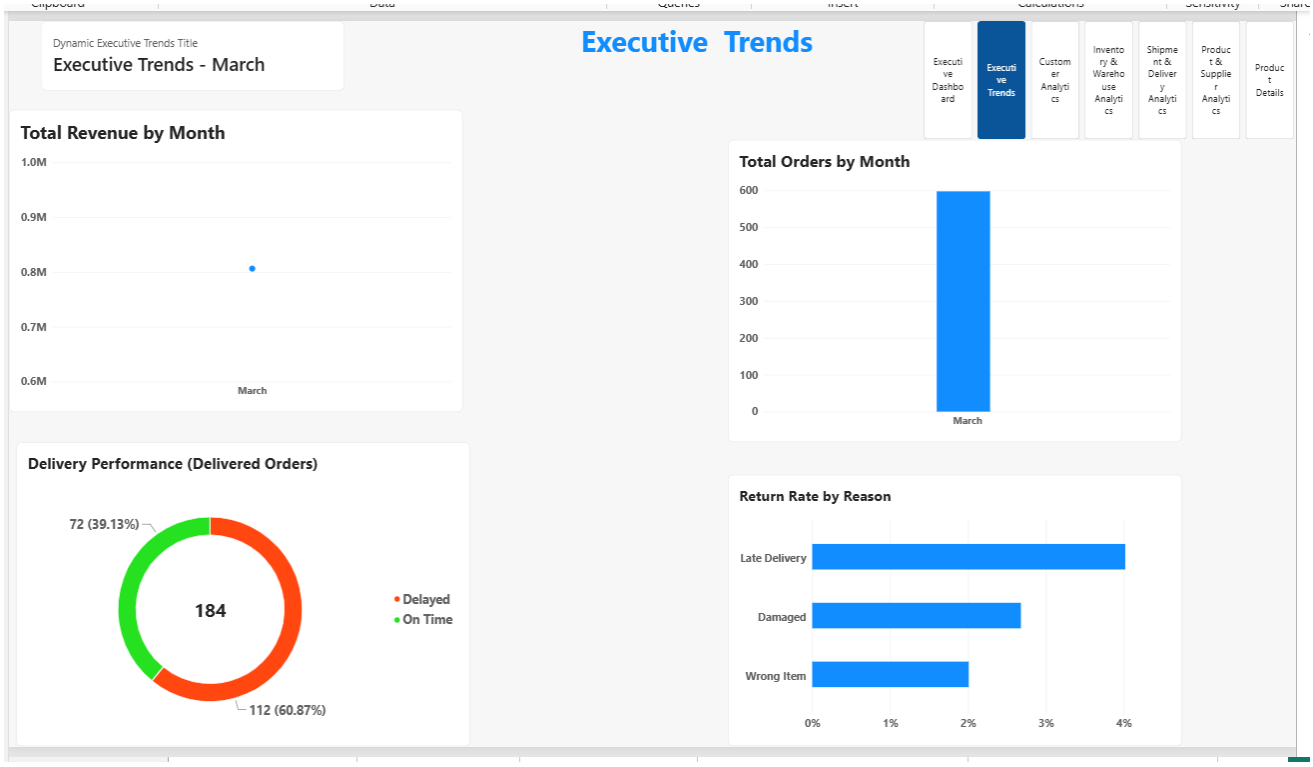


Figure 3. Executive Trends page demonstrating a dynamic month-aware title and synchronized month filtering.

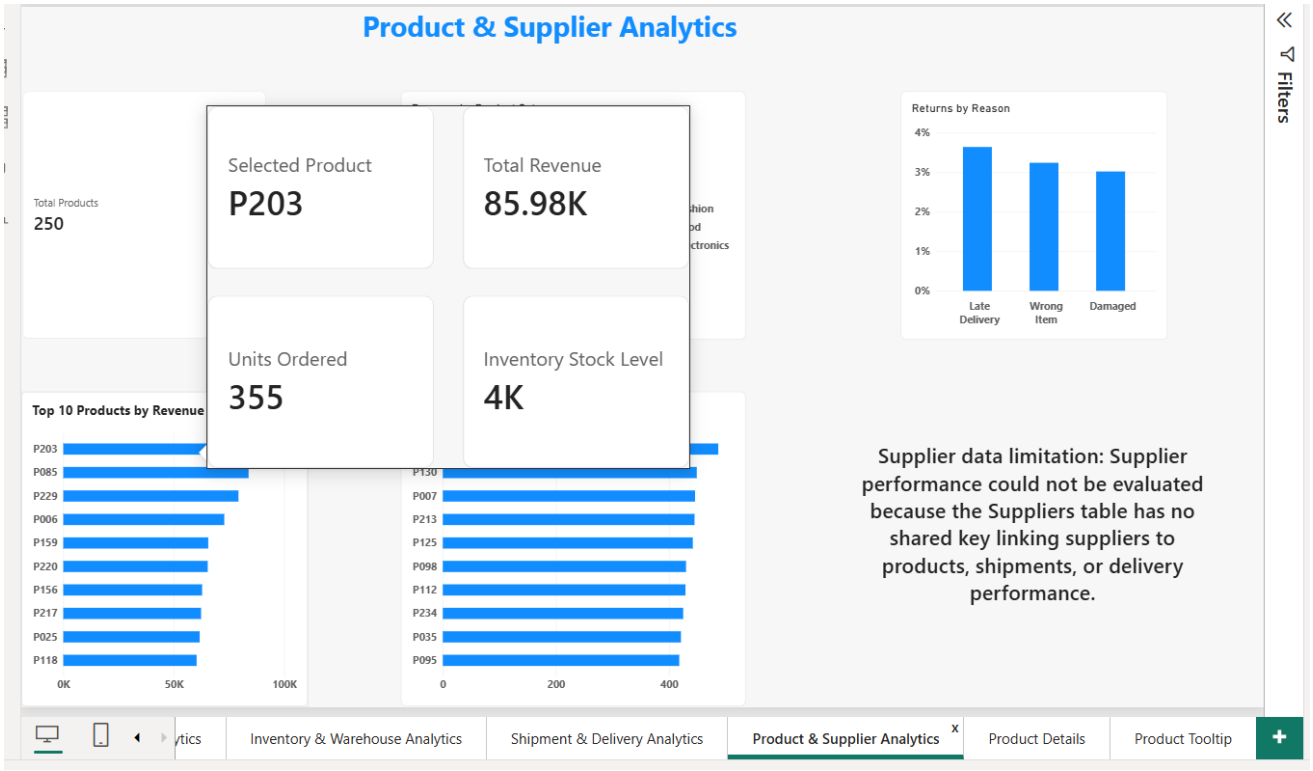


Figure 4. Product report-page tooltip showing product-specific revenue, units ordered and inventory.

5. Business Analysis & Data Storytelling

KPI snapshot

Delivered-order revenue: 7.11M | Orders: 5,000 | Shipments: 5,000 | Customers: 1,000 | Products: 250 | Average delivery delay (delivered orders): 3.52 days | On-time rate (delivered orders): 37.21%

Guide question	Evidence-based answer
1. Highest-revenue products	P203, P085 and P229 lead delivered-order revenue. P203 contributes about 85.98K.
2. Highest-revenue customers	C0223, C0773 and C0599 are the top customer revenue contributors.
3. Warehouse efficiency	W13 appears comparatively stronger on delayed-shipment volume; W01/W11 also show relatively lower delay counts. A definitive efficiency ranking is not possible without capacity/cost/throughput targets.
4. Supplier performance	Cannot be reliably evaluated because SupplierID has no relational link to operational tables.
5. Highest product return rates	Cannot be reliably calculated because Returns lacks ProductID/OrderItemID. Of 473 distinct returned orders, 379 contain multiple products, 72 contain one product, and 22 have no matching order-item row.
6. Regions with most delays	Lagos: 1,701 delayed of 2,647 shipments (64.3%); Abuja: 849/1,335 (63.6%); Kano: 636/1,018 (62.5%).
7. Peak business activity	August has the highest order volume (654). Gross order value across all statuses is about 2.85M; delivered-order revenue in August is about 997K. September is partial.
8. KPIs needing attention	On-Time Delivery Rate (~37.21%), delayed shipment volume (3,186 status records), and Average Delivery Delay (~3.52 days).
9. Operational inefficiencies	System-wide delivery delay, uneven warehouse delay volumes, low-stock exposure, and missing supplier/return relational keys.
10. Business opportunities	Protect high-value products/customers, optimize inventory allocation, improve delivery processes, prepare for peak months, and strengthen source-system data capture.

Top delivered-revenue products	Revenue
P203	85,977.99
P085	83,844.00
P229	79,260.80
P006	72,848.16
P159	65,588.75

Top delivered-revenue customers	Revenue
C0223	57,592.36
C0773	52,251.67
C0599	40,996.20
C0520	32,240.31
C0030	31,817.68

Region	Total shipments	Delayed	Delay rate
Lagos	2,647	1,701	64.3%
Abuja	1,335	849	63.6%
Kano	1,018	636	62.5%

6. Strategic Business Recommendations

#	Recommendation	Evidence / action
1	Improve on-time delivery performance	Review dispatch sequencing, route planning, handoffs and root causes; on-time performance is the clearest service-risk KPI.
2	Prioritize Lagos delivery operations	Lagos has the highest delayed volume (1,701); investigate workload, scheduling and transport bottlenecks first.
3	Benchmark stronger warehouses	Document practices in W13 and other lower-delay warehouses and test transferability to higher-delay sites.
4	Protect high-revenue products	Prioritize replenishment/service continuity for P203, P085 and P229.
5	Retain high-value customers	Use C0223, C0773 and C0599 as priority accounts for proactive service and retention actions.
6	Prepare for demand peaks	Plan inventory, staffing and delivery capacity ahead of high-volume periods such as August.
7	Rebalance inventory proactively	Use low-stock and warehouse views to trigger replenishment and inter-warehouse stock transfers.
8	Reduce returns by root cause	Track Late Delivery, Damaged and Wrong Item reasons and assign corrective actions to logistics, packaging and fulfillment teams.
9	Repair supplier and return data architecture	Add SupplierID to relevant operational tables and ProductID/OrderItemID to Returns to enable valid supplier and product-return analysis.
10	Institutionalize KPI monitoring	Review revenue, orders, on-time delivery, delay, returns and inventory KPIs on a recurring management cadence.
11	Control operating cost through exception management	Focus resources on high-delay routes/warehouses and high-risk inventory exceptions rather than broad untargeted interventions.
12	Use growth insights commercially	Coordinate high-value product availability, top-customer retention and peak-period readiness to support revenue growth.

7. Data Limitations, Controls & Interpretation

Limitation	Impact	Treatment in this project
No SupplierID link outside Suppliers	Supplier Performance Score cannot be validly calculated	Supplier limitation disclosed; no artificial relationship created
Returns lacks ProductID/OrderItemID	Product return rate is ambiguous for multi-item orders	Return analysis kept at reason/order level
No warehouse capacity/target/cost	Warehouse utilization/efficiency cannot be formally calculated	Used comparative operational indicators and disclosed limitation
No shipment/delivery timestamps	Average Delivery Time cannot be calculated	Average Delivery Delay used instead
No COGS/average inventory	Formal accounting inventory turnover unavailable	Demand-to-stock turnover proxy used and labeled

Limitation	Impact	Treatment in this project
September ends on 8 Sep	September trend is incomplete	Not compared directly with full months

Data integrity principle

Where the supplied data does not support a requested metric, the report documents the limitation rather than inventing a relationship or presenting a misleading KPI.

8. References

- AnalystLab Africa. Experience Lab Internship Programme - Official Weekly Assignment, Week 5: SwiftLog Logistics Analytics & AI Transformation Project, Data Analytics Track, Parts 1-7.
- SwiftLog Logistics supplied CSV datasets: Customers, Orders, Order Items, Products, Suppliers, Warehouses, Inventory, Shipments, Delivery Performance, Employees and Returns.